



Multimodal mixing convolutional neural network and transformer for Alzheimer's disease recognition[☆]

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ABSTRACT

Early recognition of Alzheimer's disease (AD) and its precursor state, mild cognitive impairment (MCI), is pivotal in interrupting the progression of the disease and providing suitable treatment. Recent development in deep learning techniques has drawn great research attention for improving the efficacy of AD recognition. However, numerous current methods solely utilize data from a single auxiliary domain, limiting their ability to harness valuable intrinsic insights from multiple domains. To cope with the challenge, this paper is devoted to establishing an innovative multimodal medical data fusion model, termed as MMDF, to perform Alzheimer's disease recognition. Multimodal data including clinical records and medical images are used by the proposed approach, and backbone models are constructed using various data modalities. Specifically, a vision transformer model, which is termed as MRI_ViT, is tailored to recognize AD using brain magnetic resonance imaging (MRI) data. In parallel, a novel multi-scale attention-embedded one-dimensional (1D) convolutional neural network (MA-1DCNN) is devised for analyzing clinical records. Subsequently, these basic models are combined for creating a new data fusion model to recognize Alzheimer's disease. The experimental results reveal outstanding performance compared with state-of-the-art (SOTA) methods.

1. Introduction

Alzheimer's Disease is a gradual and irreversible disease characterized by the gradual loss of nerve cells and brain tissue. It impacts a significant portion of the elderly population worldwide. Alzheimer's disease (AD) ranks as the sixth leading cause of death in the United States (Association et al., 2023). The onset of AD is usually gradual and slow, and early symptoms may be initially mistaken for a natural progression of aging or simple forgetfulness. As AD advances, it leads to a decline in cognitive functions, affecting decision-making and the ability to perform daily tasks. Presently, there is no treatment available to reverse the effects of the disease. However, following a set of recommendations may help to slow its progression (Sekhar & Jagadev, 2023). Given the absence of a cure or prevention for AD at present, the early identification of mild cognitive impairment (MCI) progression,

before irreversible brain damage occurs, holds significant importance for preventive care.

With the rising prevalence of electronic health record (EHR) systems, patients' clinical records, such as patients' demographic information, laboratory test results, neurological exams, genetic history, and other details, are accessible. Leveraging the abundant knowledge, computer-aided methods such as machine learning (ML) are increasingly employed to analyze clinical record data related to AD (Lei et al., 2023; Roncero-Parra, Parreño-Torres, Sánchez-Reolid, Mateo-Sotos, & Borja, 2024; Sasikala, 2024; Sen, Sen, & Roy, 2023; Sudharsan & Thailambal, 2023; Wang et al., 2024). However, traditional ML methods face several challenges, including reliance on manually designed features, risk of over-fitting, lack of robustness, and low accuracy. Moreover, most existing ML models in healthcare primarily rely on a single

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data modality (Huang, Pareek, Seyyedi, Banerjee, & Lungren, 2020). Although multi-modal learning (MML) has been extensively explored in natural image datasets (Hayat et al., 2020; Zellers et al., 2021) and audio-visual applications (Joze, Shaban, Iuzzolino, & Koishida, 2020), it remains relatively unexplored in the healthcare field (Hayat, Geras, & Shamout, 2022). This is mainly due to the irregular sampling and diverse formats in which patient data are stored, ranging from demographic information and laboratory results to medical images. The complexity and dynamic nature of healthcare data pose significant challenges to efficient modeling. Consequently, despite its importance, AD recognition has received limited attention in the literature due to the inherent difficulty of the task. This study aims to establish a novel Multimodal Medical Data Fusion (MMDF) model that integrates the CNN and transformer networks to allow the modeling of both medical images and clinical data-based multi-modality fusion features for AD recognition. Concretely, we devise a multi-scale attention-embedded 1D convolutional neural network, termed as MA-1DCNN, to handle clinical numerical data. The multi-scale convolution kernels are used in place of single ones in the initial convolutional layers to broaden the receptive field and augment the diversity of convolutional features. Moreover, the classical Squeeze-and-Excitation (SE) block is improved and incorporated into the network for adaptive feature calibration. We use two one-dimensional convolutional layers in place of the existing fully-connected (FC) layers to compress the model volume, and a Rectified Linear Unit (ReLU) activation function is added in front of the sigmoid layer to reduce the amount of calculation and unnecessary information loss. Conversely, for the transformer, we build a MRI_ViT sub-model that is a vision transformer fed by MRIs to train the model for extracting high-level image features. We use the MRI_ViT to identify the relationships among distinct local regions of MRIs and learn the global characteristics of images. Further, the proposed MMDF architecture integrates different sub-models based on different data modalities, e.g., the MRI data are used for training MRI_ViT and the MA-1DCNN is inferred using clinical record data. Subsequently, the various sub-models are combined, followed by two densely-connected layers with 64 and 32 neurons respectively to modify the vector dimensions. A fully connected (FC) prediction layer is then utilized for the ultimate classification of Alzheimer's disease. Briefly, the main contributions of the paper are summarized below.

- A multi-modal data-fusion model that integrates two backbone networks of the MRI_ViT and the MA-1DCNN is designed, which allows the modeling of both MRI and clinical data-based uni-modal and cross-modality fusion features.
- A novel multi-scale attention-embedded 1D convolution network architecture is elaborately devised, where multi-scale filters are substituted for the uni-scale ones and an attention mechanism is embedded into the network for clinical data-based feature extraction.
- We upgrade the classical SE block to a 1D SE block for clinical data feature calibration. Two 1×1 convolution layers are substituted for the FC layers and a ReLU function is added in front of the sigmoid layer to compress the model size while preventing unnecessary information loss.
- A self-adaptive loss function is proposed for suppressing the noise interference and mitigating the class imbalance issues, and we use this improved loss function to substitute for the traditional Cross-Entropy (CE) function.

The remainder of the paper is assembled in the following manner. Section 2 presents the related work. Section 3 elaborates on the methodology. A series of experiments are implemented in Section 4 and the performance of the model is investigated through comparative analysis. Lastly, Section 5 concludes this paper and offers recommendations for future research.

2. Related work

To perform the early detection of AD disease, Mathew, Mekkiyil, Ramasangu, Karthikeyan, and Manjunath (2016) used the Principal Component Analysis (PCA) and Discrete Wavelet Transform (DWT) for feature extraction, which were then inputted into a Support Vector Machine (SVM) to classify 71 cognitively normal (CN), 71 MCI, and 87 AD patients. Their experiments were conducted on the AD Neuroimaging Initiative (ADNI) dataset, and they achieved an accuracy of 84% and 91% for AD vs. CN and MCI vs. CN, respectively. While their findings are promising, their approach relies on manual feature extraction, heavily dependent upon domain expertise and subjective interpretation. Moradi et al. (2015) built a random forest (RF) classifier to detect early AD conversion in MCI patients. This work demonstrated the potential of their proposed method in early AD diagnosis. Nevertheless, this method can be only applied for unimodal tabular or non-imagery data, but not multi-modal data. Sudharsan and Thailambal (2023) performed the AD prediction using PCA and ML techniques. PCA is a traditional technique to reduce the dimensionality of extracted features. However, it does not account for the relationship between the dependent and independent variables. Additionally, when the number of explanatory variables is high, interpreting the extracted principal components becomes challenging (Hong, Balzano, & Fessler, 2018). Recently, deep learning (DL), especially convolutional neural networks (CNN), has shown great success in AD diagnosis as it can automatically extract features through end-to-end learning (Li et al., 2015). Helaly, Badawy, and Haikal (2021) fine-tuned a VGG19 pre-trained model to detect early AD and they achieved the best accuracy of 97% in the AD stage classification. To implement early AD diagnosis, Erdogmus and Kabakus (2023) introduced a novel CNN architecture comprised of 12 layers as a fast, cheap, yet accurate solution. Their model realized a detection accuracy of 90.4% and outperformed other state-of-the-art (SOTA) baselines. Huang (2023) established a generalizable framework based on multimodal contrastive learning and tabular attention to automatically recognize AD disease. Although it achieved a high recognition accuracy of 83.8%, their model has room for improvement and there are misclassifications, especially in distinguishing between AD and MCI, with multiple samples being misclassified as AD. Ebrahimi, Luo, Chiong, Alzheimer's Disease Neuroimaging Initiative, et al. (2021) and Qiu et al. (2020) also demonstrated that CNN could attain the best results in recognizing AD. Despite the success of CNN approaches, they require the incorporation of numerous layers to capture the relationships among features by stacking convolutional layers, thereby increasing the susceptibility to overfitting (Mostafa et al., 2021). Additionally, employing a CNN with an excessive number of layers could result in a decline in performance due to redundant information (Chen & Tsou, 2022; Wang, Li, & Wang, 2021).

More recently, Transformer has been developed for AD recognition using multi-head attention mechanism (Kumar Ravikanti & Saravanan, 2023; Lei et al., 2023; Sen et al., 2023; Wang et al., 2023). For example, Lei et al. (2023) used Transformer as the foundational network to extract multi-template region features, thereby capturing the abundant information of brain images for AD recognition. Sen et al. (2023) proposed a vision transformer and performed a comparative analysis of metaheuristic algorithms used for early AD detection. Unlike CNNs, transformers can naturally infer the relationships among features spanning long distances or distinct local regions via a self-attention mechanism. However, compared with the CNNs, transformers often struggle to infer the characteristics of local regions due to their inherent inductive bias, necessitating larger datasets for effective pre-training (Khan et al., 2023; Seydi & Sadegh, 2023; Woo et al., 2023). Additionally, most of the aforementioned methods only focus on the research of AD recognition with unimodal data such as images or numerical data, while ignoring the analysis of multi-modal data relevant to AD. To mitigate the research gap, we establish a multi-modal medical data fusion architecture that integrates the CNN and transformer networks for allowing the modeling of both medical image and clinical data-based multi-modality fusion features.

3. Methodology

In this section, we begin by describing the basic components used, followed by a detailed explanation of the proposed model.

3.1. CNN classification architecture

Since Krizhevsky, Sutskever, and Hinton (2012) first introduced the basic principles of CNN to the image classification task in 2012, CNN has obtained great attention and shown promising performance across various tasks. It can automatically learn feature representations and perform classification. The distinct characteristic of CNN is its hierarchical architecture. The CNN architecture mainly consists of a blend of convolutional layers, pooling layers, activation functions, and fully-connected layers. A convolutional layer, abbreviated as CONV, applies a filter or kernel to an input data matrix to produce an output feature map that captures certain features or patterns from the input data.

The operation of convolution is defined as

$$h_k^l = \phi(\sum_m W_{m,k}^l h_m^{l-1} + b_k^l), \quad (1)$$

where l indexes the number of layers, k refers to the index of output feature maps, input h_m^{l-1} denotes the m th feature map at $l-1$ layer, output h_k^l is the k th feature map at l th layer, W is a convolutional weight, b is the bias term, and $\phi(\cdot)$ is a activation function. A pooling layer conducts down-sampling operations by preserving only the maximum or average value within its local receptive field while discarding the rest. Similar to a convolutional layer, it operates locally. The formula is expressed as

$$Z_l = f_p(h_k^l, s), \quad (2)$$

where $f_p(\cdot)$ represents the down-sampling function, and s is the pooling size. The pooling layer is utilized for extracting a combination of features and reducing the number of parameters. Various functions like Softmax, ReLU, Sigmoid, and Tanh are used as the activation function, which helps models learn complex patterns and serve as a decision-maker (Nwankpa, Ijomah, Gachagan, & Marshall, 2018). It is noteworthy that the size of the input volume remains unchanged in this layer. The last layer is an FC layer used at the end of the network for classification prediction tasks. The FC layer can be defined by

$$h^l = \phi(W^l h^{l-1} + b^l), \quad (3)$$

where h^l is the output of the l th layer, $\phi(\cdot)$ denotes the element-wise activation function. In recent years, researchers have developed various CNN architectures, including AlexNet, LeNet, VGG, DenseNet, GoogLeNet, and more.

3.2. Transformer classification architecture

Transformer is a typical neural network that has capability to handle any data type. It was first proposed by Vaswani et al. (2017) for addressing machine translation tasks, achieving great performance in the natural language processing (NLP) domain. Unlike traditional CNNs that depend on convolutional layers, the transformer is a deep neural network structured around an encoder-decoder architecture, predominantly leveraging the self-attention mechanism for capturing intrinsic features. Given the self-attention mechanism's capability to model information across varying lengths, transformers and their derivatives have excelled in diverse tasks across multiple fields. Recently, the transformer has also been adapted to computer vision tasks and it mainly has two forms in image classification: pure self-attention framework exemplified by vision transformer (ViT) (Dosovitskiy et al., 2020) and a combined framework of CNN and self-attention represented by BoTNet-50 (Srinivas et al., 2021). A notable advantage of the transformer model lies in its robust global information extraction capability, addressing

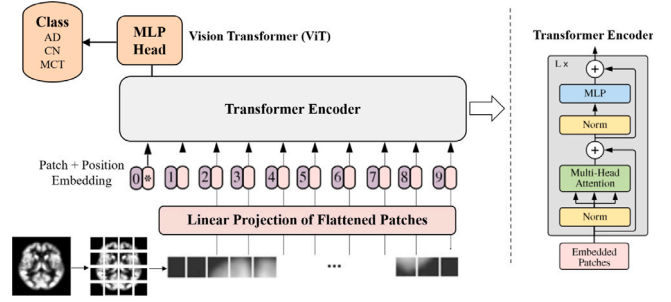


Fig. 1. The framework of the MRI_ViT.

limitations inherent in a CNN model. Nevertheless, transformer models come with certain limitations, such as losing specific local characteristics (Wang et al., 2023). Besides, though transformer is widely applied in diverse vision related tasks, it has been rarely explored in medical imaging due to larger sizes, data noises, and a substantial amount of data needed to train a usable transformer model (Dai, Gao, & Liu, 2021).

In the paper, we propose a magnetic resonance imaging (MRI)-based ViT (MRI_ViT) that efficiently leverages the MRI of brain scans for establishing the multi-modal data fusion model. Fig. 1 portrays the overall architecture of MRI_ViT. Briefly, the proposed framework comprised of a linear embedding layer, a transformer encoder block, and a final multilayer perceptron (MLP) head classifier. First, an image X with the size of $h \times w \times c$ (h , w , and c represent the height, width, and the channel number of an image) is split into patches and converted into sequences through flattening and embedding operations. Each patch is treated as a separate token by the transformer. Each patch with dimension $p \times p \times c$ is extracted from the image, which forms a patch sequence $(x_1, x_2, \dots, x_n)$ with the length of n ($n = wh/p^2$). Subsequently, the patch sequence is fed into multiple multi-head attention layers to generate the final representation. Note that a smaller patch size results in a longer sequence and vice versa. At last, the first token sequence is passed through a softmax layer for final classification. In this study, three types of data are classified, so the MLP head layer output of the transformer model is changed to three categories, including AD, CN, and MCI.

3.3. SE-block attention mechanism

Attention mechanisms can enhance DL models by concentrating on relevant information and ignoring irrelevant noise. SE block (Hu, Shen, & Sun, 2018), which adaptively recalibrates feature weights by explicitly modeling interdependencies between channels, is a widely used attention mechanism. The core operations of SE-block consist of squeeze, excitation, and reweight, in which the squeeze operation can be defined by

$$v_k = f_{sq}(u_k) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_k(i, j), \quad (4)$$

where f_{sq} represents the squeeze operation, v_k is the k th statistic value, u_k is the k th feature of the input matrix, $H \times W$ implies the spatial dimensions of u_k . Excitation implements a series of FC, ReLU, FC, and Sigmoid operations to capture the dependency relation among channels, thereby generating the feature weights, as expressed by

$$s = f_{ex}(W, v) = \sigma(A(W, v)) = \sigma(W_2 \phi(W_1 z)), \quad (5)$$

where $\sigma(\cdot)$ and $\phi(\cdot)$ refer to the sigmoid and ReLU activation functions. $W_1 \in R^{r \times K}$ and $W_2 \in R^{K \times \frac{K}{r}}$, where r is a shrinking ratio. Note that W_1 and W_2 are inferred using two non-linear FC layers in the classical SE

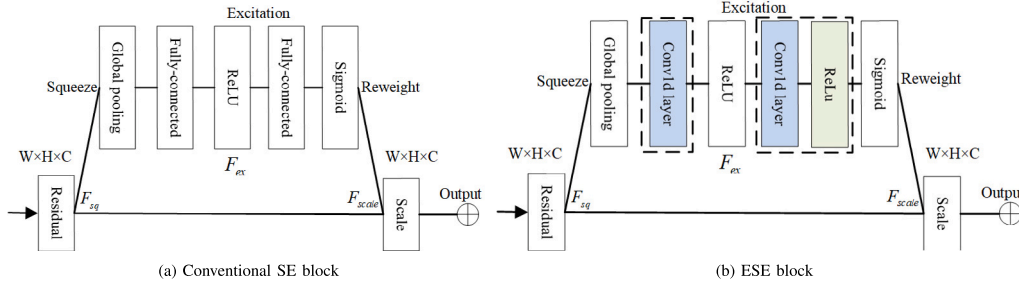


Fig. 2. Conventional SE block and ESE block.

block. After that, the reweight operation implements the recalibration for features, expressed as

$$\tilde{x}_k = f_{scale}(s_k, u_k) = s_k \cdot u_k \quad (6)$$

where $f_{scale}(s_k, u_k)$ indicates the element-wise multiplication between the scalar (s_k) and the feature map u_k . Due to the multiple parameters in fully connected (FC) layers, we replace the existing FC layers in the excitation process with two one-dimensional convolution layers. This reduces the network's volume and enhances the SE block's sensitivity. Additionally, the ReLU activation function is simple to compute and avoids gradient vanishing or saturation issues. Considering the extensive exponential calculations required by the sigmoid function alone, and the potential information loss when its derivative reaches zero, we place a ReLU activation function before the sigmoid layer. This reduces the computational load and prevents unnecessary information loss. Fig. 2 displays the difference between conventional SE block and enhanced SE (ESE) block.

3.4. Proposed approach

In the paper, we consider different data modalities, including medical image data and clinical record data (demographics, medical examination, laboratory test results, etc.) for Alzheimer's disease recognition. According to the different data modalities, we propose a Multimodal Medical Data Fusion (MMDF) architecture that combines two backbone networks of the MRI_ViT and the MA-1DCNN, allowing the modeling of both MRI and clinical data-based unimodal and cross-modality fusion features. First, sub-models based on different data modalities are built, e.g., the medical image data are used for training MRI_ViT to extract favorable image features, and the MA-1DCNN is trained using clinical tabular data. Subsequently, the sub-models are combined using a concatenate layer to form a unified data fusion model, which is then followed by two densely-connected layers with 64 and 32 neurons to modify the vector dimensions. A fully connected layer is then utilized for the ultimate recognition of Alzheimer's disease. Concretely, the detailed modeling processes are described as follows.

First, for the medical image data, we preprocess them into 2D MRIs and the image patches are created. The 2D MRIs are converted into sequences of 1D patch embeddings, which is similar to implementing word embedding in the field of NLP. In our study, the original input 2D MRI patch x of size $h \times w \times c$ is reshaped into a sequence of N patches x_p , where $N = \frac{hw}{p^2}$, p is the patch size, assigned as 6×6 in this paper.

Subsequently, the patch is projected into a vector with dimension D , to which a positional embedding of identical dimension is added, thereby incorporating sequential information into the embedding. Following this, the image is represented as a 1D embedding vector and passed through the transformer encoder block, with multi-head self-attention (MSA) serving as its central operation. All patch embeddings are normalized by a layer-normalization (LN) layer, and one key (K), query (Q), and value (V) are attained by each embedding along with the attention values gained by the MSA layer. The MSA and MLP are the two main subcomponents of the transformer encoder block.

The MLP is a fully connected feed-forward dense block, comprising two fully connected layers separated by a Gaussian error linear units (GeLU) activation function. Both subcomponents utilize residual skip connections and are preceded by a LN layer. Ultimately, the class tokens within the z_l^0 sequence are forwarded to the external classification layer to classify the categories of MRI patches. The process of deriving the final output feature from the input embeddings can be expressed using the following Eqs. (7)–(9).

$$z_l' = MSA(LN(z_{l-1})) + z_{l-1}, l = 1, \dots, L \quad (7)$$

$$z_l = MLP(LN(z_l')) + z_l', l = 1, \dots, L \quad (8)$$

$$y = LN(z_l^0) \quad (9)$$

As previously noted, a pivotal component of the transformer encoder is the MSA block. Within this block, the self-attention (SA) layer determines attention weights through the computation of scaled dot product attention across multiple heads. Each head computes the scaled dot product attention between query and key values to infer attention weights. The formula of the SA output can be expressed by

$$Attention(V, K, Q) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (10)$$

where three matrices including key matrix K , query matrix Q , and value matrix V are taken as the input of the SA block, d_k is the output dimension of the linear layer, and $\sqrt{d_k}$ provides a proper normalization to make the gradient more stable. The element-wise multiplication operation is first conducted between the element of K vector and that of Q vector, and the results determine the correlation of split patches in the MRIs. Then, the results are first scaled and sent to a softmax layer. Subsequently, the output from the softmax layer is multiplied by the V vector to identify the split patch with a high attention score.

For clinical records, a multi-scale attention-embedded 1D convolutional neural network (MA-1DCNN) is elaborately devised to extract valuable information. The dimensions of the input data are expanded so that the data set can meet the requirements of 1D convolution operation. Next, convolution operations are conducted with kernel sizes of 32, 64, and 128, respectively. During this stage, Batch Normalization (BN) and max pooling (MAP) are applied following the convolutional operations. After the max pooling layer, an attention mechanism is integrated into the network to achieve adaptive feature calibration. Precisely, an enhanced 1D SE block is embedded after the max Pooling layer to highlight the meaningful information while ignoring unwanted noises.

Moreover, the limited receptive field of a single-scale filter results in relatively uniform extracted information, thereby restricting the predictive performance of the model to some extent. To overcome this limitation, our study integrates multi-scale convolution kernels into the foundational convolutional layer of MA-1DCNN. This strategy expands the convolutional receptive field and enhances the diversity of convolutional features. Multi-scale convolutions are applied to the intermediate feature maps using filter sizes of 1, 3, and 5. Following

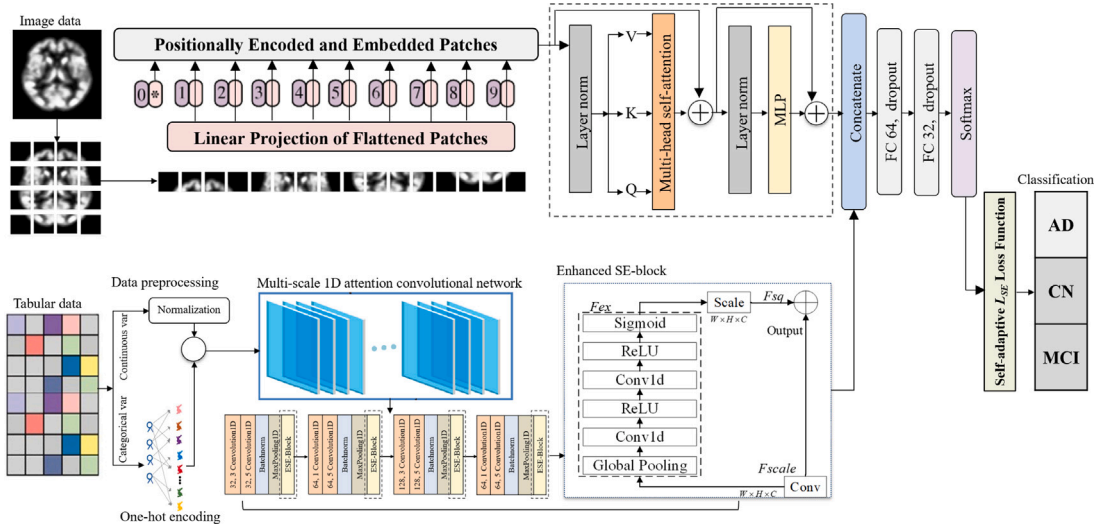


Fig. 3. The overall architecture of the proposed MMDF.

Table 1

The major parameters of the proposed network.

Module (layer)	Input shape	Output shape	Param no.	Repeat
Input-1 (InputLayer)	(None, 200, 200, 3)	(None, 200, 200, 3)	–	1
Data-augmentation	(None, 200, 200, 3)	(None, 72, 72, 3)	7	1
Patches	(None, 72, 72, 3)	(None, None, 108)	–	1
PatchEncoder	(None, None, 108)	(None, 144, 64)	16 192	1
LayerNormalization	(None, 144, 64)	(None, 144, 64)	128	17
MultiHeadAttention	(None, 144, 64)	(None, 144, 64)	66,368	8
Add layer	[(None, 144, 64), (None, 144, 64)]	(None, 144, 64)	–	16
Dense layer	(None, 144, 64)	(None, 144, 128)	8, 320	16
Input-2 (InputLayer)	[(None, 94, 1)]	[(None, 94, 1)]	–	1
conv1d (Conv1D)	(None, 94, 1)	(None, 94, 32)	128	4
conv1d-1 (Conv1D)	(None, 94, 32)	(None, 94, 32)	5152	4
BatchNormalization	(None, 94, 32)	(None, 94, 32)	128	4
MaxPooling1D	(None, 94, 32)	(None, 47, 32)	–	4
ESE-block	(None, 47, 32)	(None, 47, 32)	128	4
Multiply layer	(None, 6, 64) (None, 1, 64)	(None, 6, 64)	–	1
Flatten layer	(None, 6, 64)	(None, 384)	–	1
Dense layer	(None, 384)	(None, 3)	1155	1
Model-1 (Functional)	(None, 200, 200, 3)	(None, 3)	167 843	1
Model-2 (Functional)	(None, 94, 1)	(None, 3)	21,659,594	1
Concatenate	[(None, 3), (None, 3)]	(None, 6)	–	1
Dense layer	(None, 6)	(None, 64)	448	1
Dropout (drop = 0.1)	(None, 64)	(None, 64)	–	1
Dense layer	(None, 64)	(None, 32)	2080	1
Dropout (drop = 0.1)	(None, 32)	(None, 32)	–	1
Dense (Softmax)	(None, 32)	(None, 3)	99	1

two consecutive convolutional layers, a sequence of operations including BN layer, ReLU layer, MAP layer, and ESE block is repeated four times. These operations not only reduce computational complexity but also alleviate internal covariance shift within the model. As a result, the backbone network of our proposed MA-1DCNN comprises eight convolutional layers, with the number of convolution kernels set at 32, 64, 128, and 64. The respective kernel sizes are 3, 5, 1, 5, 3, 5, 1, and 5. Fig. 3 portrays the architecture of the proposed MMDF, with the major parameters summarized in Table 1.

To train the model, the cross entropy (CE) loss function is a frequently-used loss function in deep learning models (Mao, Mohri, & Zhong, 2023), defined as

$$L_{CE} = - \sum_{c=1}^C q(c|x) \log p(c|x), \quad (11)$$

where C is the index of the class number, $p(c|x)$ symbolizes the prediction probability of a specific sample x belonging to class c , and considering the ground-truth class y , the $q(c|x) = 0$ for all $c \neq y$ and $q(c|x) = 1$ for $c = y$. Nevertheless, the CE loss function is prone to

over-fitting to noisy labels on some categories while under-fitting on some other categories. To alleviate this challenge, Ref. Wang et al. (2019) recommended a Symmetric Cross Entropy (SCE) loss function that improves CE loss function symmetrically through a noise-robust counterpart Reverse Cross Entropy (RCE), expressed as

$$L_{RCE} = - \sum_{c=1}^C p(c|x) \log q(c|x) \quad (12)$$

$$L_{SCE} = L_{CE} + L_{RCE} \quad (13)$$

Additionally, in the case of imbalanced class samples, the performance of the CE loss function is suboptimal because it treats positive samples and negative samples as the same weight. Therefore, to overcome the category imbalance issue, Lin, Goyal, Girshick, He, and Dollár (2017) introduced a Focal-Loss (FL) function that assigns different weights to negative and positive samples. The formula of the FL function is written by

$$L_{FL} = -\omega_c (1 - p(c|x))^{\gamma} \log(p(c|x)), \quad (14)$$

Table 2
Demographic details for different diagnosis groups.

ID	Study	Male	Female	Age	Total
1	AD	164	114	75.01 ± 6.58	278
2	MCI	378	276	75.03 ± 6.59	654
3	CN	304	214	74.95 ± 6.47	518

where ω_c and γ separately denote the hyperparameters of weighting factors and modulation factors. However, the classical FL function is designed to deal with binary classification problems. For this reason, we enhance the FL function to make it suitable for multi-class problems. The enhanced FL (EFL) function can be formulated as

$$L_{EFL} = - \sum_{c=1}^C \omega_c (1 - p(c|x))^{\gamma} q(c|x) \log(p(c|x)) \quad (15)$$

$$\omega_c = \text{count}(x) / \text{count}(x \in c) \quad (16)$$

Furthermore, based on the above analysis, a hybrid loss function that leverages the merits of the EFL and SCE loss functions is considered in our scheme. Nevertheless, only the result of one loss function can be updated in the process of backpropagation, so a joint loss function must be defined to integrate the two different loss functions, and the weighted sum method is the most used scheme, which can be expressed as

$$L_{SE}^{(t)} = \sum_{i=1}^n w_i^{(t)} L_i^{(t)} = w_1^{(t)} L_{SCE}^{(t)} + w_2^{(t)} L_{EFL}^{(t)}, \quad (17)$$

where w_i denotes the weight for the i th loss function L_i , and t means the t th epoch of training. The performance of the system is related to the defined weights between losses, while tuning these weights by hand is a challenging task and consumes significant costs. Therefore, we include the loss weights in the definition of the loss function itself and develop a self-adaptive way to update loss weights through callback, which manages the changes internally. The loss weight update can be defined as

$$w_i^{(t+1)} \leftarrow w_i^{(t)} - \lambda \nabla_{w_i} L_{grad}, \quad (18)$$

where λ is a constant hyper-parameter, and L_{grad} denotes the gradient loss, which is introduced especially to depict the loss caused by loss weight w . The formula of gradient loss L_{grad} is written by

$$L_{grad}(t, w_i^{(t)}) = \sum_i \left| G_i^{(t)} - \overline{G^{(t)}} \times [r_i^{(t)}]^{\alpha} \right|_1, \quad (19)$$

$$G_i^{(t)} = \left\| \nabla_{\theta} w_i^{(t)} L_i^{(t)} \right\|_2, r_i^{(t)} = \frac{L_i^{(t)} / L_i^{(0)}}{E_{task}[L_i^{(t)} / L_i^{(0)}]} \quad (20)$$

Among them, $G_i^{(t)}$ is the value of gradient normalization on the i th loss function in the t th epoch of training, which is calculated by the L2 norm of the weighted loss gradient. $\overline{G^{(t)}}$ is the mean of gradient normalization for all losses in the t th epoch of training. $r_i^{(t)}$ implies the relative training speed of the i th loss in the t th epoch of training. Overall, the loss weight is treated as an optimization parameter in this solution, and the L_{grad} of loss weight w is established in each epoch of update. The initial weights of the L_{SCE} and L_{EFL} are both set to 0.5, and the gradient update is implemented for each epoch of training.

4. Experiment analysis

To evaluate the effectiveness of our proposed approach, we conduct experiments using the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset. State-of-the-Art (SOTA) Alzheimer's disease recognition methods are chosen for comparative study.

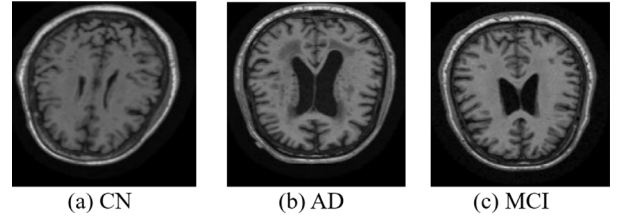


Fig. 4. The MRI samples of ADNI dataset.

4.1. Data description

The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early Alzheimer's disease (AD). The multi-modal data, including demographic, clinical, biochemical biomarker, neuropsychological, and MRI, are available in the ADNI dataset for analysis by investigators. Among them, the brain MRI is classified into CN, MCI, and AD groups. To accelerate the training and testing of the models, the three-dimensional brain scans have been sliced into two-dimensional (2D) slice structures along the coronal axis by taking the central brain images. Thus, all the brain MRI images have been used to train a deep learning model for the multi-class AD/MCI/CN classification applied to evaluate model performance. This main table ADNIMERGE consolidates key variables from different case report forms and biomarker laboratory summaries across the ADNI protocols, including ADN11, ADN10, and ADN12. A total of 116 variables, such as Participant roster ID (RID), Participant ID (PTID), Visit code (VISCODE), and more, are included in this table. It is worth noting that the unused columns like FLDSTRENG, FSVERSION, Ventriles, and the null columns are excluded from the original data. The categorical variables, including PTGENDER (Female, Male), PTETHCAT (Hisp/Latino, Not Hisp/Latino, Unknown), PTRACCAT (Asian, Black, White), and PTMARRY (Married, Never Married, Divorce, Unknown), are converted into dummy/indicator variables. Our goal is to develop an advanced deep learning model that can efficiently fuse multi-modal data to determine and classify Alzheimer patients into AD, MCI, and CN categories. Fig. 4 displays the representative brain scans in the ADNI dataset, and demographic details of different diagnosis groups are presented in Table 2.

4.2. Experimental setup

The experiments are conducted using Python 3.6, utilizing widely-used libraries such as TensorFlow, Keras, Scikit-learn, and Matplotlib, and are accelerated by a graphics processing unit (GPU). The hardware environment for operating the Python deep learning framework to implement Alzheimer's disease recognition includes AMD EPYC™ 7502P 32-Core processors, 32 GB RAM, and an NVIDIA RTX A6000 GPU, among other components, used for algorithm execution. The design, training, and testing of the model are all conducted on a Linux-based operating system.

4.3. Evaluation metrics

We evaluate the performance of models using multiple metrics including Accuracy (Acc), Precision (Pre), Recall (Rec), and F1-Score ($F1$). Among these metrics, Acc is mastered by the superiority of TP and TN over the total number of samples, Pre is mastered by the superiority of TP overall positive outcome, $Recall$ is mastered by the superiority of TP over TP and FN , and $F1$ is mastered by the

Table 3

The training accuracy of network models.

Models	Training for 30 epochs				Training for 100 epochs				
	Training Acc %	Validation Acc %	Training loss	Validation loss	Training Acc %	Validation Acc %	Training loss	Validation loss	Run time (m)
MLP	91.05	84.41	0.2610	0.2854	94.24	88.82	0.1082	0.1278	00:00.41
1D-CNN	97.64	91.10	0.0654	0.2848	99.88	94.61	0.0056	0.1450	00:04.08
Transformer	89.57	86.41	0.1387	0.2816	90.78	88.82	0.0762	0.2024	00:12.25
V-CNN	96.69	92.06	0.0013	0.0029	99.52	96.76	0.0002	0.0018	00:05.57
Proposed MMDF	97.72	92.94	0.0008	0.0037	99.70	97.35	0.0002	0.0019	00:12.45

superiority of $2*TP$ over $2*FP$ and false outcome. The formulas of these metrics can be mathematically expressed by

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (21)$$

$$Precision = \frac{TP}{TP + FP} \quad (22)$$

$$Recall = \frac{TP}{TP + FN} \quad (23)$$

$$F1 - Score = \frac{2TP}{2TP + FP + FN} \quad (24)$$

where TP implies the true positive that is the number of positive samples accurately recognized. FN is the false negative, which denotes the number of negative samples incorrectly recognized. FP represents false positives, indicating the number of incorrectly identified positive samples. TN stands for true negatives, representing the number of negative samples correctly identified.

4.4. Baselines

To demonstrate the robustness of the proposed model, four SOTA network methods including Multilayer Perceptron (MLP) classifier, one-dimensional CNN (1D-CNN), Transformer (Vaswani et al., 2017), and a VGG-style CNN (V-CNN) architecture, are chosen for comparison study. Moreover, the most used ML methods, including k-nearest neighbors (k-NN), SVM, Random Forest (RF), and light gradient boosting machine (lightGBM), are also selected to compare models. Among them, the V-CNN is similar to our method, but it uses a VGG-style CNN instead of the MRI_ViT module to implement MRI-based Alzheimer's disease recognition. The sizes of the convolutional kernel in V-CNN are 32, 64, and 128, respectively. 1D-CNN is a one-dimensional convolutional network architecture comprised of 32, 64, and 128 convolutional kernels. The 1D-CNN method is similar to the MA-1DCNN sub-model of our method, but it does not introduce attention mechanism and uses uni-scale convolutions with uniform filter size of 3. Different from the proposed MMDF and V-CNN that can fit multi-modal data, other methods can only take a single data modality like numerical data as the input. Therefore, these compared methods are conducted on the clinical record (tabular) data of the ADNI dataset. The core hyperparameters of benchmark models are set to the same as that of the proposed method. Specifically, we set the mini-batch size to 16, with a learning rate (lr) of 1×10^{-3} , conduct training for 100 epochs, and employ Adam as the optimizer (Kingma & Ba, 2014).

4.5. Results and analysis

The proposed MMDF is employed to perform the AD recognition on the ADNI dataset. The dataset is randomly divided into the training, validation, and test sets in a ratio of 7:2:1. Various augmentation techniques, including random flipping, rotation, illumination, and zooming, are implemented on the brain MRIs to enhance the diversity and variability of training images in the dataset. Brain images containing symmetric characteristics are ideally used for flipping, which presents the network from favoring characteristics, e.g., the diseases are not shown on one side of the organ, but might have been presented on the other side. The rotation operation is implemented with a factor of

0.02, and the zooming is operated with width and height factors of 0.2, respectively. Table 3 summarizes the training performance of all the methods. As seen in Table 3, the proposed approach consistently outperforms other SOTA methods. After 30 and 100 training epochs, the validation accuracy of our method reaches 92.94% and 97.35%, respectively, marking the highest accuracy among all algorithms. Thereafter, the trained model is used to recognize Alzheimer's disease on test data. The prediction results are summarized in Table 4. Fig. 5 displays the receiver operating characteristic (ROC) curve of the recognition results by the proposed method, and the test confusion matrices of different methods are portrayed in Fig. 6.

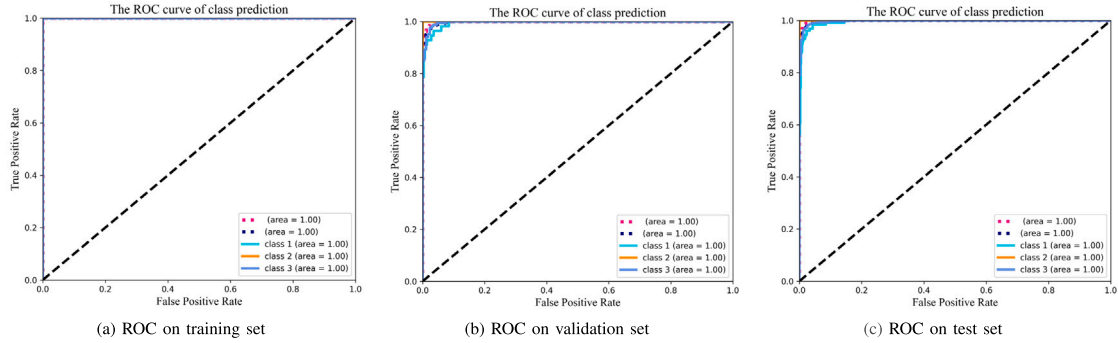
It can be observed from Table 4 that the proposed method has attained the Accuracy of 97.35% and 97.65%, the Recall of 96.62% and 96.40%, and the F1-Score of 96.45% and 96.69% on the validation set and test set, respectively, which achieve the best accuracy of all algorithms. Also, it can be visualized from Fig. 5 that the AD recognition results show optimal operating points on the ROC curves, and the curves of all categories are aggregated to the left top corner of the graph, which indicates the effectiveness of the proposed approach. Similar results can be revealed by the confusion matrix of Fig. 6(i), where the majority of samples in each category are correctly identified by the proposed method. As a consequence, the average accuracy of the proposed method reaches 97.65% on the test dataset, which indicates that the proposed MMDF has a capability for recognizing Alzheimer's disease. In addition, the average running time-consuming of the MMDF is 12 min and 45 s. Though the time consumption is slightly higher than that of the compared methods, it does not pose a challenge with current hardware levels. The main explanation is that the MMDF architecture integrates the backbone networks of MA-1DCNN and MRI_ViT, which realizes the modeling of unimodal and cross-modal fusion features from tabular and image data. In the MA-1DCNN, the multi-scale filters are substituted for the uni-scale ones in the underlying convolution layers, which enlarges the receptive fields and boosts the richness of the convolutional features. Simultaneously, the incorporated ESE attention block makes the networks focus on meaningful features while ignoring useless noises. Moreover, multi-modal data are fused in our modeling scheme, thereby improving model accuracy. The MRI_ViT which is an architecture of MRI-based vision transformer comprised of multiple MSA is used to leverage the MRIs for Alzheimer's disease recognition. Among them, the MSA is the core component and achieves impressive effects in multiple fields. Beyond that, a self-adaptive loss function L_{SE} (see Eq. (17)) is proposed for suppressing the noise interference and mitigating the class imbalance issue, which further improves the model accuracy. In contrast, the other methods either rely solely on CNNs or utilize commonly-used machine learning algorithms. Despite various parameter tuning and optimization efforts, these methods fail to achieve the desired performance. Consequently, our proposed approach achieves the highest accuracy in comparative experiments for Alzheimer's disease recognition.

Furthermore, we conducted a comparative performance analysis of our method against results from prior studies on the ADNI dataset, as shown in Table 5. It is important to note that this table only serves as a reference, given the different data handling protocols applied by these studies. Despite these differences, our findings indicate that the proposed MMDF approach yields promising results and outperforms the majority of existing methods reported in the literature. This broader comparison underscores the effectiveness of our approach and its potential advantages in Alzheimer's disease recognition.

Table 4

The prediction results of different methods.

No.	Methods	Training set (%)				Validation set (%)				Test set (%)				Time (m)
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	
1	SVM	70.13	74.75	57.25	64.84	69.41	74.66	56.19	64.12	67.91	68.55	54.59	60.78	00:00:41
2	MLP	94.24	96.45	88.55	92.33	88.82	91.80	81.67	86.44	89.81	91.21	83.89	87.40	00:00:41
3	k-NN	83.85	83.18	79.37	81.23	75.59	74.90	68.50	71.56	71.89	69.62	66.17	67.85	00:00:38
4	Random Forest	92.50	93.39	86.77	89.96	90.59	91.50	87.05	89.22	90.86	90.70	85.62	88.09	00:00:38
5	LightGBM	96.21	94.61	94.37	94.49	93.82	92.23	91.06	91.64	94.96	92.95	93.21	93.08	00:00:38
6	1D-CNN	98.45	97.12	98.38	97.75	96.17	95.28	94.24	94.76	94.61	91.94	93.51	92.72	00:04:08
7	Transformer	90.78	93.82	85.43	89.43	88.82	91.54	84.24	87.74	85.71	89.02	78.73	83.56	00:12:25
8	V-CNN	98.94	98.41	98.56	98.48	96.76	96.11	95.03	95.57	96.02	95.30	94.18	94.74	00:05:57
9	Proposed MMDF	99.93	99.84	99.95	99.89	97.35	96.27	96.62	96.45	97.65	96.98	96.40	96.69	00:12:45

**Fig. 5.** Predicted ROC curves of the proposed method.**Table 5**

Comparison with previous literature on the ADNI dataset (%).

References	Modality	Test datasets	Methods	Acc	Rec	F1
Sahumbaiev, Popov, Ramirez, Górriz, and Ortiz (2018)	MRI	185 AD, 160 CN, 185 MCI	CNN (HadNet)	88.31	–	–
Bucholt et al. (2019)	MRI, Clinical data	46 AD, 178 CN, 263 MCI	SVM	82.90	82.40	–
Abuhmed, El-Sappagh, and Alonso (2021)	MRI, Clinical data	339 AD, 419 CN, 613 MCI	Multivariate BiLSTM	86.08	88.67	87.67
Venugopalan, Tong, Hassanzadeh, and Wang (2021)	MRI, Clinical data	707 AD, 598 CN, 699 MCI	DL + RF	78.00	78.00	78.00
Qiu et al. (2020)	MRI, Clinical data	188 AD, 229 CN	FCN + CNN	96.80	95.70	96.50
Kushol, Masoumzadeh, Huo, Kalra, and Yang (2022)	MRI	159 AD, 229 CN	Transformer	88.20	95.60	–
Li et al. (2022)	MRI	197 AD, 366 CN	Trans-ResNet	93.85	90.49	–
Golovanevsky, Eickhoff, and Singh (2022)	MRI, Clinical data	35 AD, 165 CN, 39 MCI	DL + Attention (MADDi)	96.88	–	91.41
Liu et al. (2023)	MRI, Clinical data	102 AD, 137 CN, 174 MCI	Multi-Modal Mixing Transformer(3MT)	93.00	95.60	–
This study	MRI, Clinical data	127 AD, 268 CN, 459 MCI	Proposed MMDF	97.65	96.40	96.69

Table 6

The ablation experiment results.

Ablation approaches	Training set (%)				Validation set (%)				Test set (%)				Time (m)
	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	
Delete MRI_ViT	97.93	97.22	98.07	97.64	95.29	95.07	93.70	94.38	94.50	93.33	93.34	93.34	00:06:22
Delete ESE block	98.84	97.65	99.08	98.36	95.88	93.79	96.01	94.89	96.95	94.82	96.97	95.88	00:11:03
Replace multi-scale conv with uni-scale one	99.67	99.79	99.22	99.50	97.06	97.05	95.23	96.13	97.18	96.34	94.65	95.49	00:10:66
Replace L_{SE} with L_{CE}	99.73	99.62	99.60	99.61	95.88	95.04	93.64	94.33	96.72	96.47	94.11	95.28	00:12:08
Proposed MMDF	99.93	99.84	99.95	99.89	97.35	96.27	96.62	96.45	97.65	96.98	96.40	96.69	00:12:45

4.6. Ablation study

We further conduct ablation studies to evaluate the performance of the proposed approach. We separately remove the MRI_ViT module, the enhanced SE block, the multi-scale convolutions (conv), and the optimized loss function from the proposed MMDF architecture to investigate the performance of the ablation models. Table 6 summarizes the results of ablation experiments. The detailed ablation process and result analysis is summarized below.

4.6.1. Ablation on MRI_ViT

First, we estimate the effect of the MRI_ViT module for recognizing Alzheimer's disease. We freeze the MRI_ViT from the network and utilize the model without MRI_ViT for prediction. We observe a significant decrease in the recognition accuracy of the ablation model, where

the test Accuracy, Precision, and Recall drop to 94.50%, 93.33%, and 93.34% (decrease by 3.15%, 3.65%, and 3.06%), respectively. Also, the test F1-Score drops to 93.34% (decrease by 3.35%). While the recognition performance of the ablation model remains superior to that of most comparative baseline methods, it experiences a notable decline compared to the proposed MMDF. This ablation experiment suggests that the removal of MRI_ViT module significantly affects the model's performance when compared to the cumulative impact of the MRI_ViT module in our proposed method.

4.6.2. Ablation on enhanced SE block

Subsequently, we remove the enhanced 1D SE-block from the network to investigate the efficacy of the proposed method. We notice a minor accuracy drop for the ablation model where the test Accuracy and F1-Score drop to 96.95% and 95.88% (decrease by 0.70% and

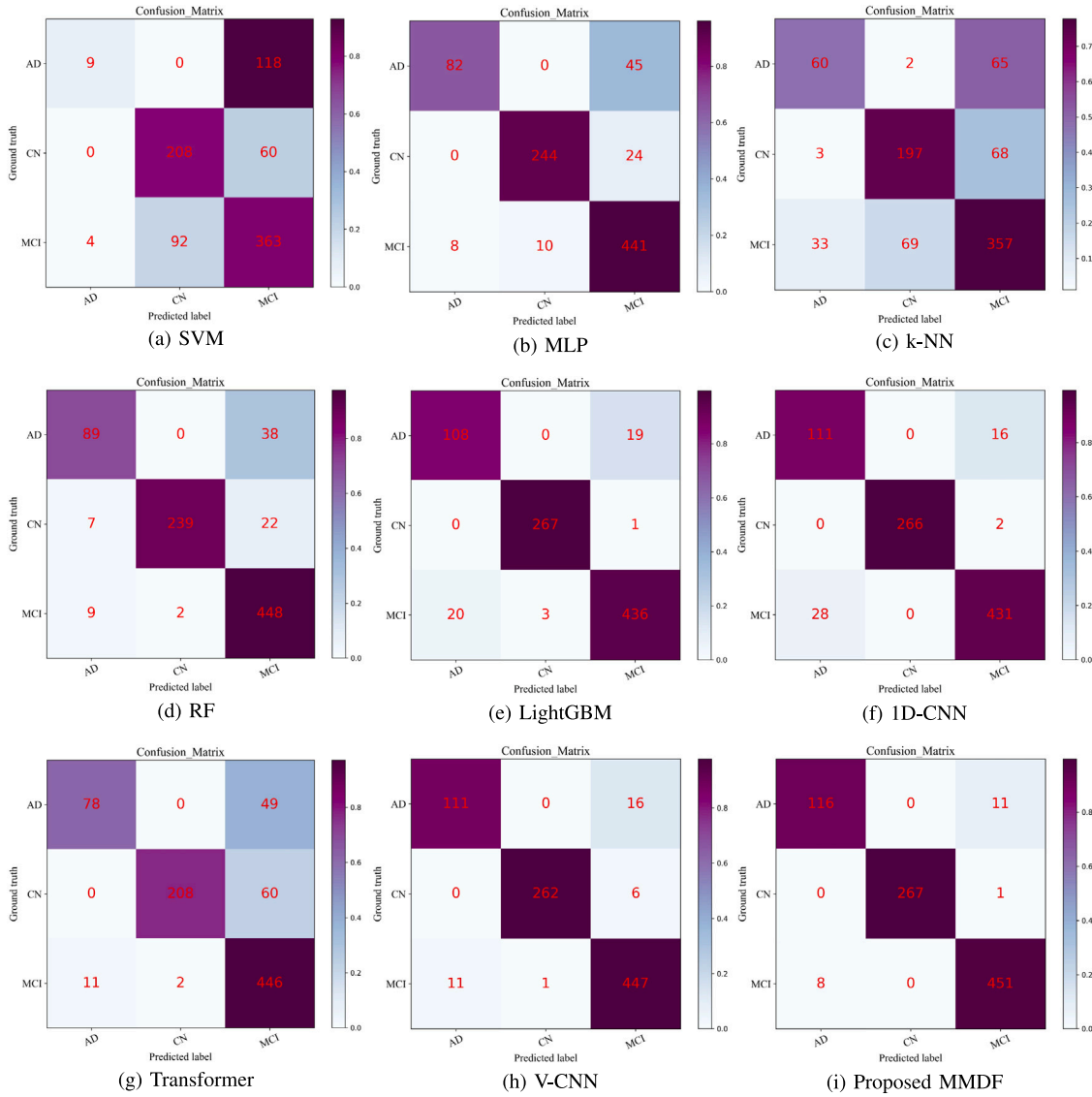


Fig. 6. The test confusion matrix of different methods.

0.81%), respectively. The *Precision* drops a bit more to 94.82% (decrease by 2.16%). The findings from this ablation experiment suggest that the improved 1D SE-block plays a significant role in boosting the performance of the proposed method.

4.6.3. Ablation on multi-scale convolutions

Next we examine the effectiveness of employing multi-scale convolutions to broaden the receptive field and enhance the diversity of convolutional features. A uni-scale filter with the size of 3 is substituted for the multi-scale ones in the networks, and we notice a slight accuracy drop in the ablation model. The test *Accuracy*, *Precision*, and *F1-Score* drop to 97.18%, 96.34%, and 95.49% (decrease by 0.47%, 0.64%, and 1.20%), respectively. The *Recall* drops a bit more to 94.65% (decrease by 1.75%). This ablation experiment confirms the superiority and feasibility of the multi-scale filters employed in the bottom convolutional layers of the MA-1DCNN.

4.6.4. Ablation on optimized loss function

Further, we evaluate the effectiveness of the optimized loss function by replacing the proposed self-adaptive L_{SE} loss function with the conventional CE loss function. In assessing the impact of the optimized loss function, we observe a noticeable decrease in the AD recognition

accuracy of the ablated model, with the test recall rate dropping to 94.11% (a decrease of 2.29%). The main reason for this significant drop in the ablated model lies in the fact that Alzheimer's disease constitutes only a small fraction of the total samples. The CE loss function fails to address the issue of imbalanced samples, thereby compromising accuracy. This ablation experiment highlights that our proposed self-adaptive loss function outperforms the conventional CE loss function typically used in DL models for Alzheimer's disease recognition.

4.7. Hyperparameter finetuning

Given that hyperparameters significantly affect the performance of learning models, we further investigate the impact of hyperparameter on the proposed model, including learning rate and batch size. The learning rate is initialized to $(l_r)_0 = 0.001$, which gradually decays to 0 following a cosine decay function. The range for the batch-size hyperparameter is defined as $(|B|) \in \{8, 16, 32, 64, 128, 256\}$. Mathematically, the cosine decay function is defined as

$$l_r_t = \frac{1}{2} \left(1 + \cos\left(\frac{t\pi}{T}\right) \right) l_{r_0}, \quad (25)$$

where t is the current step, l_r_t refers to the current learning rate, and T denotes the total number of steps at which the learning rate diminishes

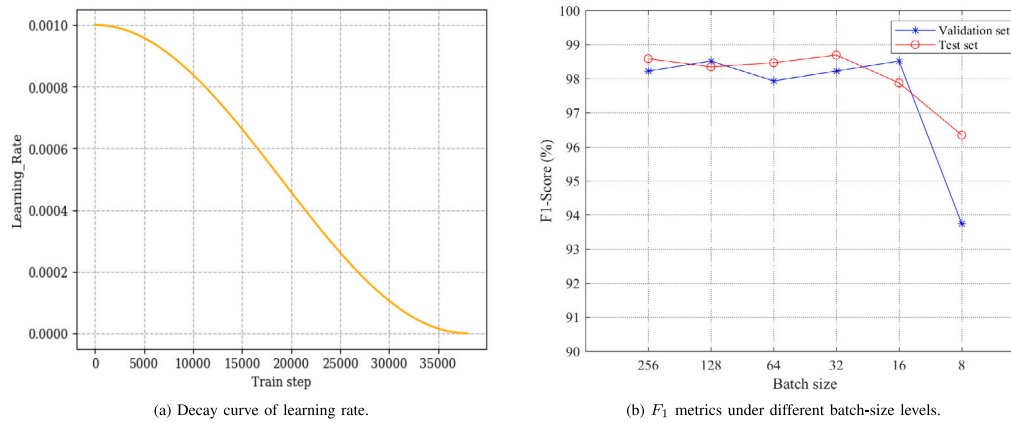


Fig. 7. Learning rate and batch-size hyperparameter fine-tuning.

Table 7

The comparison results of hyperparameter finetuning.

Batch size	Training set (%)				Validation set (%)				Test set (%)				Time (m)
	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	
256	99.93	99.84	99.95	99.89	98.23	97.99	97.53	97.76	98.59	98.19	98.21	98.20	00:05:56
128	100.00	100.00	100.00	100.00	98.52	98.60	98.04	98.32	98.36	98.20	97.49	97.84	00:12:09
64	100.00	100.00	100.00	100.00	97.94	97.42	97.33	97.37	98.47	98.28	97.76	98.02	00:17:57
32	100.00	100.00	100.00	100.00	98.23	97.99	97.53	97.76	98.71	98.60	97.68	98.14	00:29:15
16	99.80	99.76	99.82	99.79	98.52	98.19	97.81	97.99	97.89	97.51	96.69	97.10	01:09:57
8	98.06	96.03	98.70	97.35	93.52	90.59	94.81	92.65	96.25	93.76	96.95	95.33	02:26:26

to 0. Using the hyperparameters from these sets, the proposed method was trained for 100 epochs on the dataset, and Fig. 7(a) depicts the decay curve of the learning rate. Table 7 demonstrates the prediction performance with different batch size. From Fig. 7 and Table 7, it is evident that when the batch size is specified as 128, a small overall improvement is obtained on the training, validation, and test sets of the proposed method. Furthermore, a slight improvement is observed when the batch-size is set to an initial value of 16 and a dynamic learning rate is applied using cosine decay. Compared to the models without finetuning, the Accuracy and Precision rise to 97.89% and 97.51% on the test dataset, which are increases of 0.24% and 0.53%, respectively. Similarly, the Recall and F1-Score also rise to 96.69% and 97.10%, reflecting increases of 0.29% and 0.41%, over the proposed method before finetuning. While hyperparameter finetuning does require additional time, the experimental results indicate that it can lead to enhancements in performance.

5. Conclusions

Early AD diagnosis and recognition is essential in preventing and managing its progression, where data-driven and machine learning-based methods can be extremely beneficial. Clinical practices yield diverse data modalities, presenting an opportunity for integrating this information into a cohesive model for enhanced diagnostic accuracy, optimized resource allocation, and thus facilitate informed clinical decision-making. In pursuit of this goal, the paper proposes a novel multimodal medical data fusion model, noted as MMDF, which combines two backbone sub-models of the MA-1DCNN and the MRI_ViT, allowing the modeling of both clinical records and MRI data-based unimodal and cross-modality fusion features. Concretely, the MRIs of the brains are used to train the MRI_ViT sub-model and the MA-1DCNN is trained using clinical record data. Subsequently, the two backbone networks are combined to create a unified data-fusion model. Experimental results showcase the effectiveness of the proposed model, outperforming current SOTA methods. Despite its enhanced recognition accuracy, the proposed model exhibits slightly higher computational overhead. To address this, our future work will incorporate model pruning algorithms for simplification.

CRediT authorship contribution statement

Junde Chen: Methodology, Software, Validation, Data curation, Visualization, Writing – original draft. **Yun Wang:** Interpretation of data, Writing – review & editing. **Adnan Zeb:** Validation, Writing – review & editing. **M.D. Suzauddola:** Visualization, Writing – review & editing. **Yuxin Wen:** Conceptualization, Supervision, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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